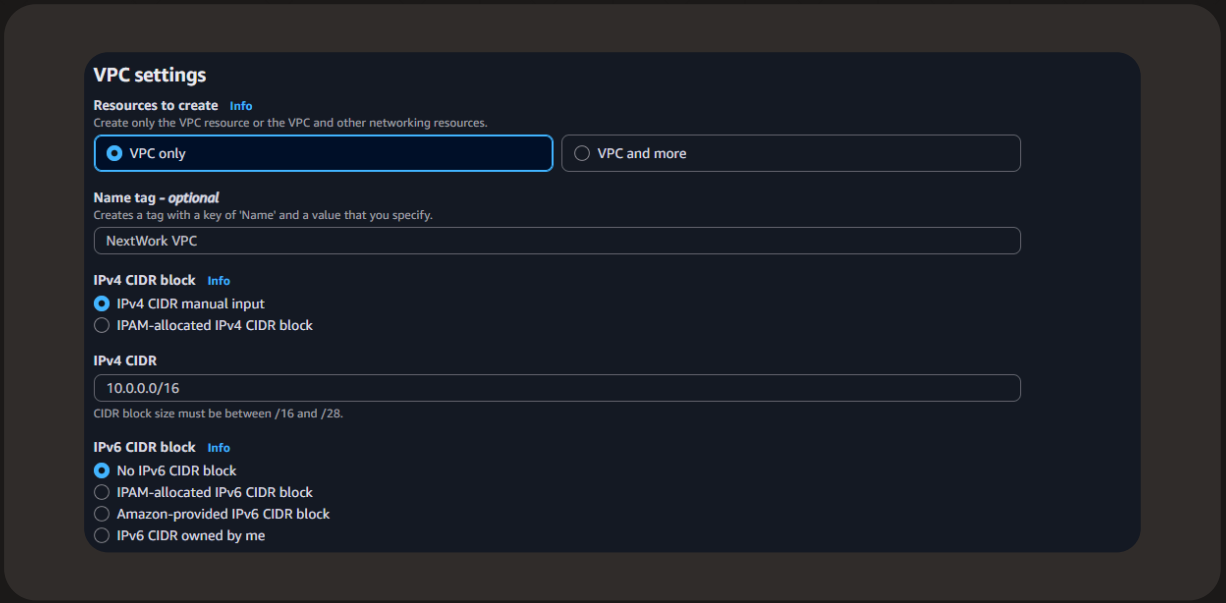
BUILD A VIRTUAL PRIVATE CLOUD

By Reynaldo Eusebio | September 2026



Introducing Today's Project!

In this project, I will demonstrate how to create a Virtual Private Cloud (VPC) in AWS. I'm doing this project to learn how to create a VPC, subnets, and an Internet Gateway.

Virtual Private Clouds (VPCs)

What I did in this step

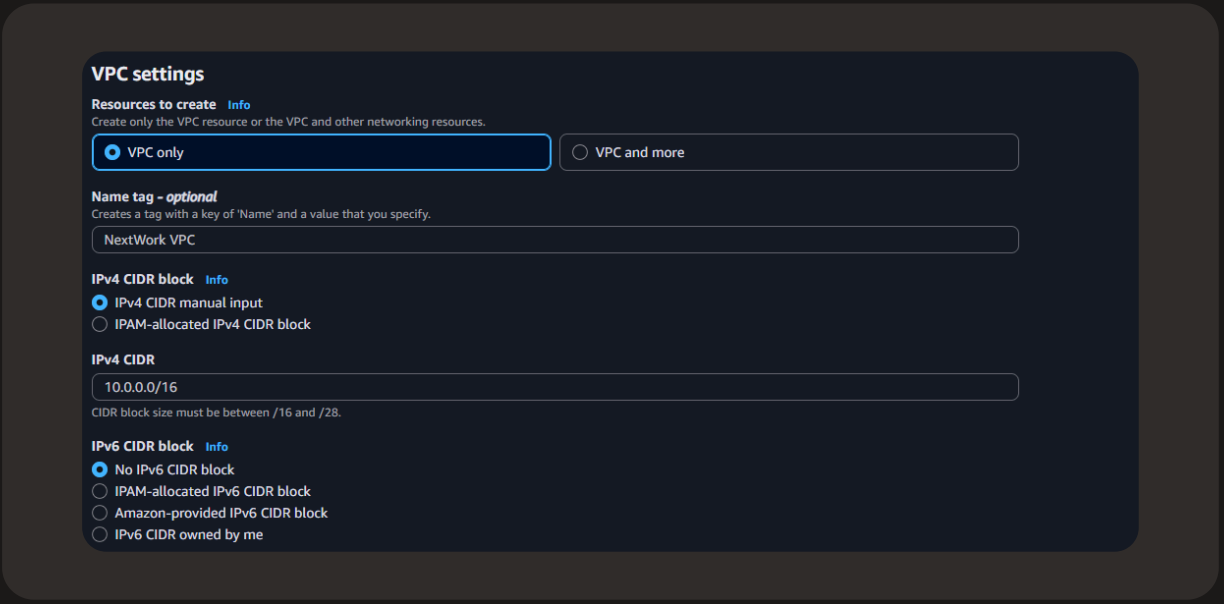
In this step, I will create a VPC first because, based on the diagram, the VPC is the outer part of the network. The VPC provides the overall network where I will create the subnets and other resources.

How VPCs work

VPCs are just like a city in a country. A VPC is the private network where I can organize and control my AWS resources. VPCs are also the reason why resources can be made private to me.

Why there is a default VPC in AWS accounts

There was already a default VPC in my account when I created my AWS account. This is because AWS automatically creates a default VPC that we can use to launch and test AWS resources, such as EC2 instances.



Defining IPv4 CIDR blocks

To set up my VPC, I had to define an IPv4 CIDR block, which is 10.0.0.0/16. This CIDR block defines the range of IP addresses that can be used within my VPC.

Subnets

What I did in this step

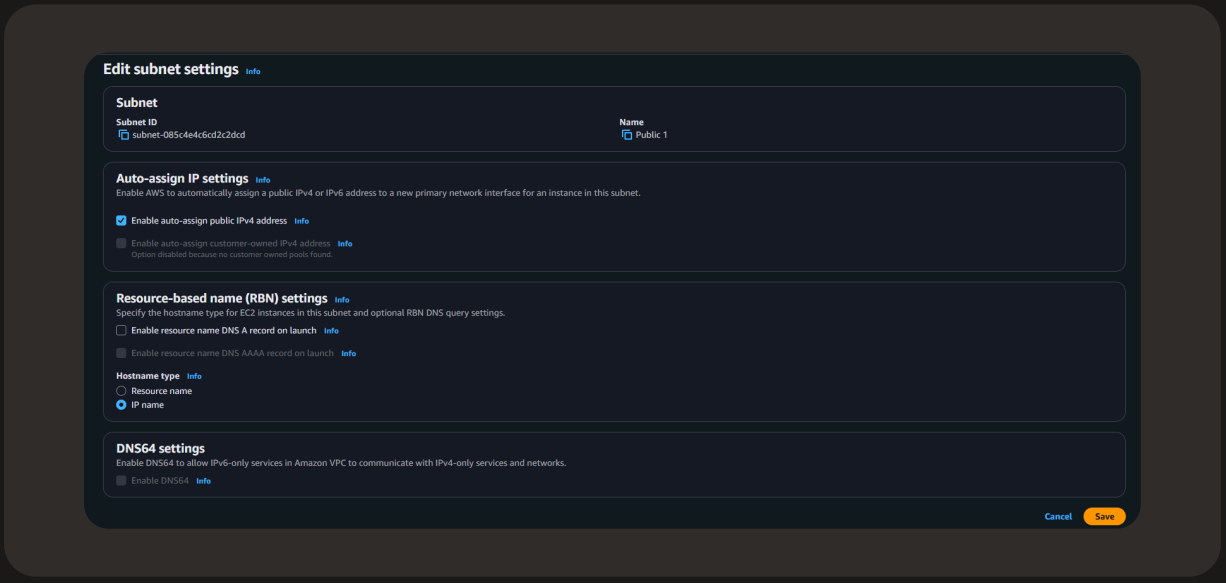
In this step, I will create a subnet because it will divide the big space of my VPC into smaller spaces or neighborhoods where I can organize my resources.

Creating and configuring subnets

Subnets are like neighborhoods inside my VPC There are already subnets existing in my account, one for every Availability Zone in my Region.

Public vs private subnets

The difference between public and private subnets is that a public subnet can connect to the internet, while a private subnet cannot connect directly to the internet. For a subnet to be considered public, its route table must have a route to an Internet Gateway.



Auto-assigning public IPv4 addresses

Once I created my subnet, I enabled the auto-assign public IPv4 address setting so that resources launched in the public subnet can automatically receive a public IP address. This allows them to use the public IP address immediately.

Internet gateways

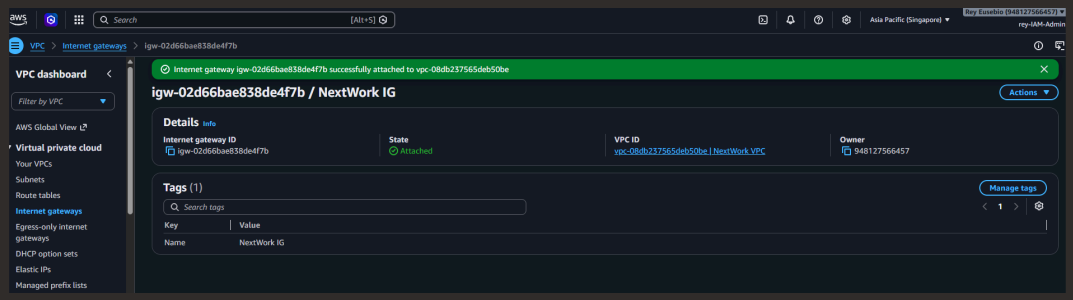
What I did in this step

In this step, I will connect my VPC to an Internet Gateway because I want my VPC to have access to the internet.

Setting up internet gateways

Internet Gateways act as a bridge that connects my VPC to the internet or the outside world.

Attaching an Internet Gateway to a VPC allows my VPC to connect to the internet. If I missed this step, my VPC would not have access to the internet.



Using the AWS CLI

What I'm doing in this extension

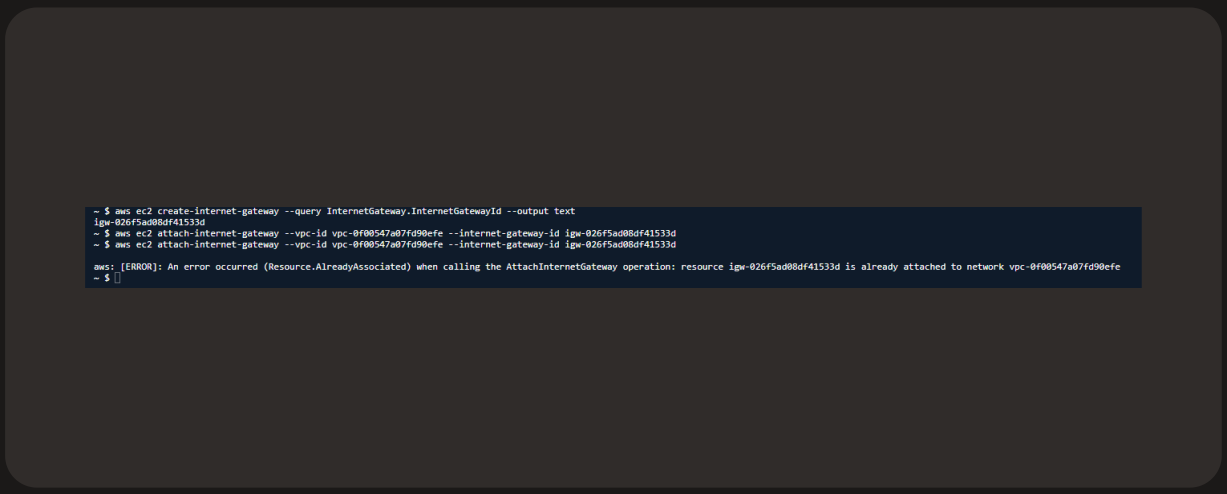
In this project extension, I will open AWS CloudShell because it is used to run commands to set up the VPC, subnet, and Internet Gateway.

Exploring CloudShell and CLI

VPC resources could also be created with CloudShell, which is a browser-based terminal. CLI is a command-line interface that allows me to create and manage AWS resources by typing commands instead of using the AWS Console.

Debugging my setup

To set up a subnet, you can use the command aws ec2 create-subnet --vpc-id VPC-ID --cidr-block ADD-CIDR-BLOCK-HERE. Make sure to include a valid CIDR block to avoid errors.



Comparing CloudShell vs AWS Console

Compared to using the AWS Console, an advantage of using commands is that I can create and manage resources faster by typing commands. An advantage of using the Console is that it provides a visual interface that makes it easier to see and understand the resources I am creating. Overall, I preferred using the AWS Console because it was easier for me to understand and navigate.